QP CODE:24900081

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATIONS NOVEMBER 2024

First Semester

MG1DSCECT100,MG1DSCECC100-EMERGING ELECTRONICS

2024 Admission Onwards

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

SCHEME OF EVALUATION

PART A

Multiple Choice Questions

Answer all questions.Each question carries 1 Mark

	Learning Domain	CO No(s)
1.Directly Proportional	**U**	**1**
2.Henry	**U**	**1**
3.All the above	**U**	**1**
4.Pulsating DC	**U**	**2**
5.2Vp	**U**	**2**
6.50Hz	**A**	**2**
7.Noise Immunity	**U**	**3**
8.Temperature	**K**	**3**
9.It must be replaced	**U**	**3**
10.Electrons	**U**	**1**

PART B

Short Answer Questions

Answer 4 questions.Each question Carries 5 Marks

11. Extract the characteristics of AC signals. AC Signal Waveform-2 Mark Characteristics-(Period,Frequency Amplitude) 3 mark	**U**	**1**
12. Interpret the color code "yellow-violet-red-gold" and determine the resistance value. 5 Mark for correct answer 4700Ω ± 5% or 4.7KΩ ± 5%	**A**	**1**
13. Interpret the significance of biometrics in security. Explanation-5 Mark	**U**	**1**
14. What is a Ripple factor?.What is its value of a Half Wave and Full Wave Rectifier? Definition-3 mark Ripple factor Half Wave rectifier-1.21 (1 mark) Ripple factor Full Wave rectifier-0.48 (1 Mark)	**U**	**2**
15. What is a clipper circuit? Explanation-3 mark	**K**	**2**

Diagram-2 mark

16. How does a Miniature Circuit Breaker (MCB) protect against short circuits?	**U**	**3**

Explanation-5 mark

PART C

Essay Questions

Answer any 2 Questions.Each carries 10 Marks

17. Relate AC and DC signals to electronic devices, explaining their applications.	**U**	**1**

Comparison of Ac and DC Signal-6 Mark

Application-4 Mark

18. Describe how a transistor operates as an amplifier, including the importance of biasing and its applications in electronic devices.	**U**	**2**

Amplifier Working with diagram-5mark

Importance of Biasing-3 Marks

Application-2 Mark

19. Describe the working of a center-tapped full-wave rectifier, including the role of each component and its advantages over a half-wave rectifier.	**A**	**2**

Working -4 Mark

Diagram-4 Mark

Advantage-2 mark

20. What is an LDR? Describe the applications of Light Dependent Resistors (LDRs) in real-world devices and systems.	**K**	**3**

Explanation-7 Mark

Applications-3 Mark